

Please check the examination details below before entering your candidate information

Candidate surname		Other names <i>Nethuka</i>	
Centre Number		Candidate Number	

Pearson Edexcel International GCSE

Wednesday 5 November 2025

Morning (Time: 2 hours)	Paper reference 4MA1/1H
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Mathematics A
PAPER 1H
Higher Tier



You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator. Tracing paper may be used.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain NO credit.

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL TWENTY EIGHT questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency (f)	Mid-Value (x)	fx
$0 \leq K < 5$	8	$\frac{0+5}{2} = 2.5$	20
$5 \leq K < 10$	7	$\frac{10+5}{2} = 7.5$	52.5
$10 \leq K < 15$	3	$\frac{10+15}{2} = 12.5$	37.5
$15 \leq K < 20$	10	$\frac{15+20}{2} = 17.5$	175
$20 \leq K < 25$	2	$\frac{20+25}{2} = 22.5$	45
$\Sigma f = 30$		$\Sigma fx = 330$	

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

$$\begin{aligned}
 \text{Estimated Mean (m)} &= \frac{\Sigma fx}{\Sigma f} \\
 &= \frac{330}{30} \\
 &= \underline{\underline{11 \text{ km}}}
 \end{aligned}$$

11 kilometres

(Total for Question 1 is 4 marks)



P 7 9 7 8 5 A 0 3 2 8

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{211^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

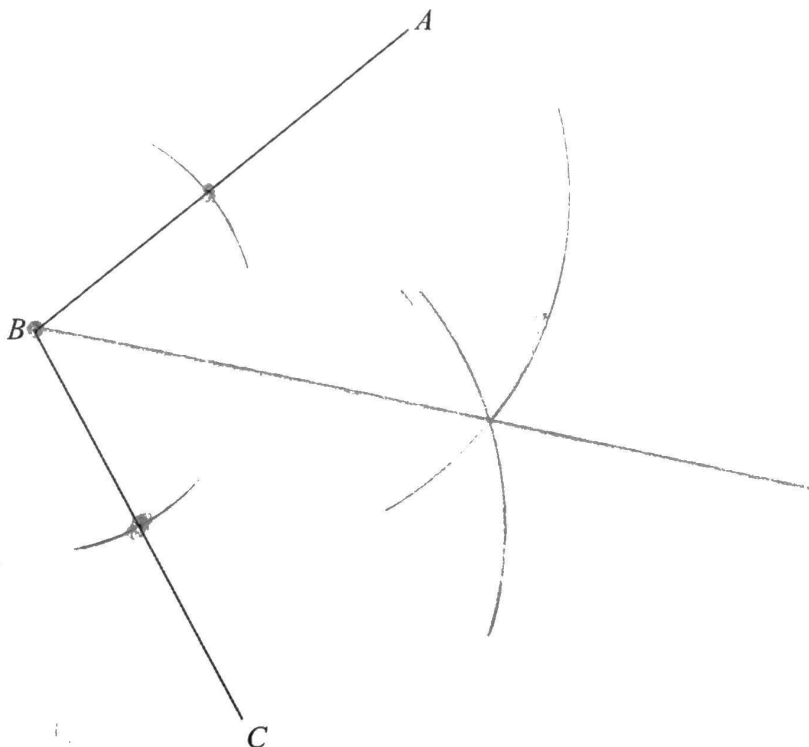
$$\frac{2^2 \times 60}{\sqrt{9}}$$

$$= \underline{\underline{80}}$$

80

(Total for Question 2 is 2 marks)

- 3 Using ruler and compasses only, construct the bisector of angle ABC . Show all your construction lines.



(Total for Question 3 is 2 marks)



- 4 e, f and g represent integers such that

e f 7 g 12 17

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

range

$$17 - e = 15$$

$$e = 17 - 15 = \underline{2}$$

mean:

~~$$2 + f + 10 + 12 + 17 = 51$$~~

~~$$6 \times 2 + f + 7 + 10 + 12 + 17 = 9 \times 6$$~~

median:

~~$$2 \times \frac{7+9}{2} = 8.5 \times 2$$~~

$$7+9 = 17$$

$$9 = 11 - 7 = \underline{10}$$

$$f + 48 = 54$$

$$f = 54 - 48$$

$$f = \underline{6}$$

$$e = \underline{2}$$

$$f = \underline{10}$$

$$g = \underline{6}$$

(Total for Question 4 is 3 marks)

- 5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

$$3 \times \frac{26}{3 \times 7} + \frac{5 \times 7}{3 \times 7}$$

$$\frac{78}{21} + \frac{35}{21}$$

~~$$\frac{113}{21}$$~~

$$\therefore \frac{113}{21} = 5\frac{8}{21}$$

(Total for Question 5 is 3 marks)

6 Eli and Peta share \$275 in the ratio 2:3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

$$\begin{array}{c} E \quad P \\ 2:3 \\ 2+3 = 5 \end{array}$$

$$\left. \begin{array}{l} \text{Total} \\ \text{Savings for} \\ \text{charity} \end{array} \right\} = 30 + 52.8 = \underline{\underline{\$82.8}}$$

Eli:

$$\frac{2}{5} \times 275 = \$110$$

charity Savings $\Rightarrow \frac{3}{11} \times 110 = \30

Peta:

$$\frac{3}{5} \times 275 = \$165$$

charity Savings $\Rightarrow 0.32 \times 165 = \52.8

$$= \$82.8$$

(Total for Question 6 is 4 marks)

The weight of a bag of apples is 475 g correct to the nearest g

$$\frac{1}{2} = \pm 0.5$$

(a) Write down the lower bound of the weight.

$$LB = 475 - 0.5 = \underline{\underline{474.5}}$$

$$474.5 \text{ g} \\ (1)$$

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

$$\frac{10}{2} = \pm 5$$

$$UB = 120 + 10 = \underline{\underline{130}}$$

$$130 \text{ cm} \\ (1)$$

(Total for Question 7 is 2 marks)



8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$$\frac{8^{(-2+9)}}{8^{10}} = 8^n$$

$$\frac{8^7}{8^{10}} = 8^n$$

$$\frac{n}{7-10} = n$$

$$n = -3$$

$$\underline{\underline{n = -3}}$$

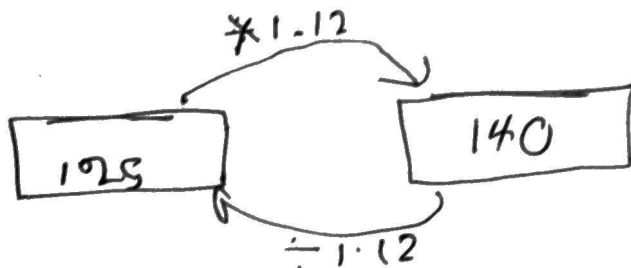
(Total for Question 8 is 2 marks)

- 9 Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.

$$\frac{100 + 12}{100}$$

Work out the number of points that Hari scored playing the game on Monday.

$$= 1.12$$



$$\therefore \text{Monday score} = \frac{140}{1.12}$$

$$= \underline{\underline{125 \text{ points}}}$$

125 pts

(Total for Question 9 is 3 marks)



P 7 9 7 8 5 A 0 7 2 8

Turn over ►

10 (a) Simplify $(6m)^3$

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(1)

(b) Solve $5y - 20 < 7y - 1$

$$20 - 1 < 7y - 5y$$

$$\frac{19}{2} < \frac{2y}{2}$$

~~19~~

$$\frac{19}{2} < y$$

$$\frac{19}{2} < y$$

(c) Factorise fully $15w^2x^3 + 25w^3x^2$

$$\begin{array}{r} 3 \overline{) 15} \\ 5 \end{array}$$

$$\begin{array}{r} 5 \overline{) 25} \\ 5 \end{array}$$

$$\begin{aligned} 15 &= 3 \times 5 \\ 25 &= 5 \times 5 \end{aligned} \quad (2)$$

$$\underline{\underline{5w^2x^2(3x^3 + 5w)}}$$

$$5w^2x^2(3x^3 + 5w)$$

(2)

(d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$

$$\frac{3 \cdot 12a^5c^7}{12a^2c^3}$$

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(2)

11 ABC is a right-angled triangle.

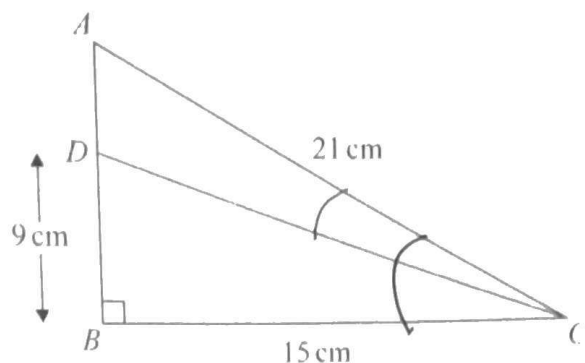


Diagram NOT
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

Pythagoras theorem,

$$AC^2 = AD^2 + BC^2$$

$$21^2 = AD^2 + 15^2$$

$$AD^2 = 21^2 - 15^2$$

$$AD = \sqrt{21^2 - 15^2} = 6\sqrt{6}$$

$\triangle ACB$

$$\tan \theta = \frac{6\sqrt{6}}{15}$$

$$\theta = \tan^{-1}\left(\frac{6\sqrt{6}}{15}\right)$$

$$\theta = 44.415$$

$$\begin{aligned} \therefore \hat{ACD} &= \hat{ACB} - \hat{DCB} \\ &= 44.415 - 30.963 \\ &= 13.452 \\ &= \underline{\underline{13.5^\circ}} \end{aligned}$$

$\triangle DCB$

$$\tan \theta = \frac{9}{15}$$

$$\theta = \tan^{-1}\left(\frac{9}{15}\right)$$

$$\theta = 30.963$$

$$13.5^\circ$$

(Total for Question 11 is 4 marks)



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12

Diagram NOT
necessarily drawn

The diagram shows a trapezium with parallel sides of length 10 cm and 14 cm.

The area of the trapezium is 117 cm².

Find the height.

$$9x + 21 + 2x + 21 = 117 \quad (\text{Area of Trapezium})$$

$$9x + 2x + 21 + 21 = 117$$

$$11x + 42 = 117$$

$$11x = 117 - 42$$

$$\frac{11x}{11} = \frac{75}{11}$$

$$x = 7$$

$$\text{Let } h = 14 \quad (9 \times 14 + 21) = 117$$

$$\text{Area} = \frac{\text{height} \times \text{sum of parallel sides}}{2}$$

$$117 = \frac{h}{2} \times 14$$

$$\frac{h}{2} = \frac{117 \times 2}{14}$$

$$h = \frac{234}{14}$$

(Total for Question 12 is 4 marks)

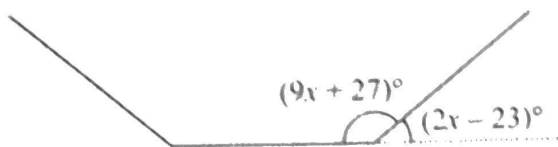


Diagram NOT
accurately drawn

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n

$$9x + 27 + 2x - 23 = 180 \quad (\text{as in straight line})$$

$$9x + 2x + 27 - 23 = 180$$

$$11x + 4 = 180$$

$$11x = 180 - 4$$

$$\frac{11x}{11} = \frac{176}{11}$$

$$x = 16$$

$$\therefore \text{Interior } \angle = 9 \times 16 + 27 = 171$$

$$1 \text{ Interior } \angle = \frac{\text{Int } \angle}{n}$$

$$n \times 171 = \frac{360}{2} \times n$$

$$\frac{360}{171} = \frac{171n}{171}$$

$$n = \frac{360}{171}$$

(Total for Question 12 is 4 marks)

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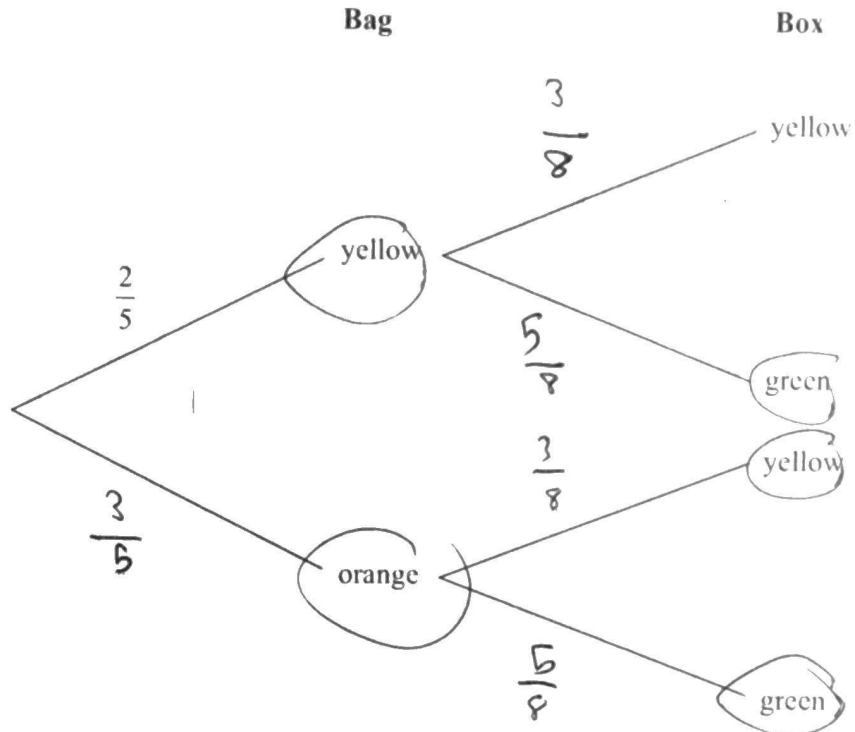
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- (a) Complete the probability tree diagram.



(2)

- $$p(y, g) + p(0, y) + p(0, g)$$

$$\left(\frac{2}{5} \times \frac{5}{8}\right) + \left(\frac{3}{5} \times \frac{3}{8}\right) + \left(\frac{3}{5} \times \frac{5}{8}\right)$$

$$= \frac{1}{4} + \frac{9}{40} + \frac{1}{8}$$

$$\frac{17}{20}$$

(3)

(Total for Question 13 is 5 marks)

14 (a) Expand and simplify $7x(3x+2)(2x-5)$

~~340~~

$$7x[3x(2x-5) + 2(2x-5)]$$

$$7x[6x^2 - 15x + 4x - 10]$$

$$7x[6x^2 - 11x - 10]$$

$$= \underline{42x^3 - 77x^2 - 70x}$$

$$42x^3 - 77x^2 - 70x$$

(3)

(b) Solve $\frac{7x-9}{7x-2y} + \frac{5-2y}{7-2y} = 5$

Show clear algebraic working.

$$1 \times \frac{63 + 10y}{14y} = 5 \times 14y$$

$$63 + 10y = 70y$$

$$63 = 70y - 10y$$

$$\frac{63}{60} = \frac{60y}{60}$$

$$\underline{y = 1.05}$$

$$y = 1.05$$

(3)

(Total for Question 14 is 6 marks)



- 15 AOB is a sector of a circle with centre O and radius 12 cm



Diagram NOT
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

$$\begin{aligned}
 \text{Area of Sector} &= \frac{\theta}{360} \times \pi r^2 \\
 &= \frac{75}{360} \times \pi \times 12^2 \\
 &= 30\pi \\
 &= 94.247 \\
 &= \underline{\underline{94.2\text{ cm}^2}}
 \end{aligned}$$

94.2 cm²

(Total for Question 15 is 2 marks)

- 16 Ivan asked 15 people how many books they read last year.

Here are his results.

1, 1, 3, 4, 6, 7, 8, 9, 10, 12, 15, 25, 30, 37, 50

Work out the interquartile range of the number of books read.

$$IQR = Q_3 - Q_1$$

Q_3

$$\frac{3}{4}(n+1)$$

$$= \frac{3}{4}(15+1) = 12^{\text{th}} = 25$$

Q_1

$$\frac{1}{4}(n+1)$$

$$= \frac{1}{4}(15+1) = 4^{\text{th}} = 4$$

$$IQR = Q_3 - Q_1$$

$$= 25 - 4$$

$$= \underline{\underline{21}}$$

21

(Total for Question 16 is 2 marks)



- 17 Show that $\frac{2\sqrt{7}+2}{\sqrt{7}-3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

$$\begin{aligned} & \frac{2\sqrt{7}+2}{\sqrt{7}-3} \times \frac{(\sqrt{7}+3)}{(\sqrt{7}+3)} \\ & \frac{2\sqrt{7}(\sqrt{7}+3) + 2(\sqrt{7}+3)}{7-9} \\ & \frac{2 \times 7 + 6\sqrt{7} + 2\sqrt{7} + 6}{-2} \\ & \frac{14 + 6 + 6\sqrt{7} + 2\sqrt{7}}{-2} \\ & = \frac{20 + 8\sqrt{7}}{-2} \end{aligned}$$

$$\begin{aligned} & \frac{20 + 8\sqrt{7}}{-2} \\ & -2(-10 - 4\sqrt{7}) \end{aligned}$$

(Total for Question 17 is 3 marks)

- 18 Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$

$$p = \sqrt{\frac{y+w}{3y-t}}$$

$$3y-t \times p^2 = \frac{y+w}{3y-t} \times 3y-t$$

$$3yp^2 + tp^2 = y + w$$

$$3yp^2 - y = w - tp^2$$

$$y(3p^2 - 1) = \frac{w - tp^2}{3p^2 - 1}$$

$$y = \frac{w - tp^2}{3p^2 - 1}$$

$$y = \frac{w - tp^2}{3p^2 - 1}$$

(Total for Question 18 is 4 marks)

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19 The diagram shows triangle ABC

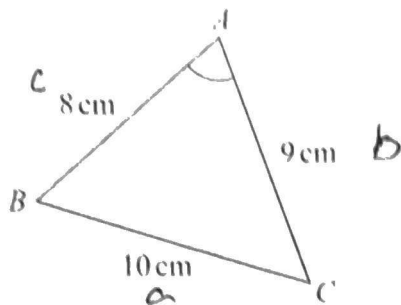


Diagram NOT
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC
Give your answer correct to one decimal place.

cos rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{9^2 + 8^2 - 10^2}{2 \times 9 \times 8}$$

$$A = \cos^{-1}\left(\frac{5}{16}\right)$$

$$A = 71.710$$

$$\underline{\underline{A = 71.8^\circ}}$$

$$71.8^\circ$$

(Total for Question 19 is 3 marks)

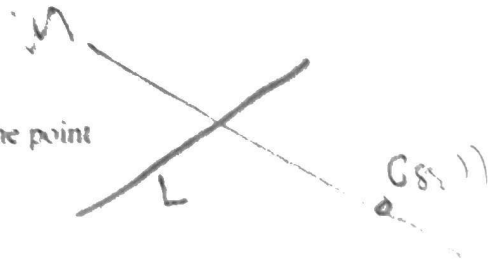


20 The straight line L has equation $y = 4x - 7$

The straight line M is perpendicular to L and passes through the point with coordinates (8, 1)

Find an equation for M

Give your answer in the form $y = mx + c$



$$m_1 \times m_2 = -1$$

$$4 \times m_2 = -1$$



$$m = -\frac{1}{4} = -\frac{1}{4}$$

$$M \rightarrow y = -\frac{1}{4}x + c$$

$\therefore$ substitute (8, 1) to find c

$$1 = -\frac{1}{4} \times 8 + c$$

$$1 = -2 + c$$

$$c = 1 + 2 = 3$$

$$\therefore y = -\frac{1}{4}x + 3$$

$$y = -\frac{1}{4}x + 3$$

(Total for Question 20 is 3 marks)

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21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

$$5 \left[x^2 - 4x + 4.6 \right]$$

$$5 \left[\left(x + \frac{b}{2} \right)^2 + c - \left(\frac{b}{2} \right)^2 \right]$$

$$5 \left[\left(x + \frac{-4}{2} \right)^2 + 4.6 - \left(\frac{-4}{2} \right)^2 \right]$$

$$5 \left[(x - 2)^2 + 4.6 - 4 \right]$$

$$5 \left[(x - 2)^2 + 0.6 \right]$$

$$\underline{5(x - 2)^2 + 3}$$

$$5(x - 2)^2 + 3$$

(Total for Question 21 is 3 marks)

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

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23 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$$\frac{2 \left(\frac{5-2y}{3y} \right) + 5}{1 - \left(\frac{5-2y}{3y} \right)}$$

$$\frac{10 - 4y + 5}{6y}$$

$a =$

(Total for Question 23 is 3 marks)

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(-1 , 3)
(1)

(ii) $y = f(x) + 7$

(4 , 10)
(1)

(iii) $y = f(2x)$

(2 , 3)
(1)

(Total for Question 24 is 3 marks)

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- 25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

$$2x^2 - 7x + 8x - 56 > 0$$

$$x(2x - 7) + 4(2x - 7)$$

$$(x + 4)(2x - 7)$$

$$x = -4 \text{ or } x = \frac{7}{2}$$



$$x < -4 \text{ or } x > 3.5$$

$$x < -4 \text{ or } x > 3.5$$

(Total for Question 25 is 3 marks)

Turn over for Question 26

- 26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

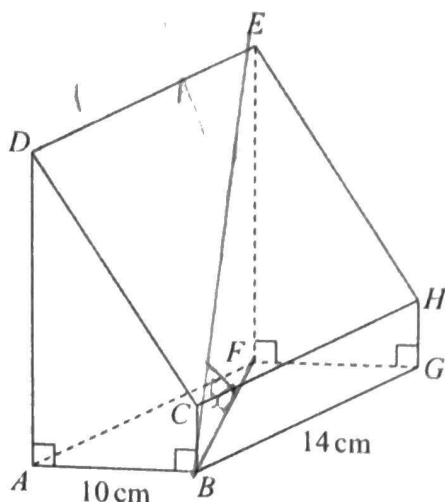


Diagram NOT
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

$$\text{Volume of Prism} = \text{C.S Area} \times l$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

$$AB = FB = 10 \text{ (Opposite sides of rectangle)}$$

$$\text{since } ABGF \rightarrow \square \quad FGB = 90^\circ$$

$\therefore$ Pythagoras

$$FB^2 = BG^2 + FG^2$$

$$FB = \sqrt{14^2 + 10^2} = 2\sqrt{74}$$

$$\therefore \tan 50 = \frac{EF}{BF}$$

$$\tan 50 = \frac{EF}{2\sqrt{74}}$$

$$EF = 2\sqrt{74} \times \tan 50$$

$$EF = 20.503$$

$$\therefore EF = DA = 20.503$$

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$$\therefore PC = \frac{1}{5} \times 20.503$$

$$= 4.1006$$

C.S Area

$$\text{Area of trapezium} = \frac{h(a+b)}{2}$$

$$= \frac{10(20.503 + 4.1006)}{2}$$

$$= 123.018$$

$$\therefore \text{Volume} = 123.018 \times 14$$

$$= 1,722.252$$

$$= \underline{\underline{1720}}$$

1,720 cm³

(Total for Question 26 is 5 marks)

Turn over for Question 27

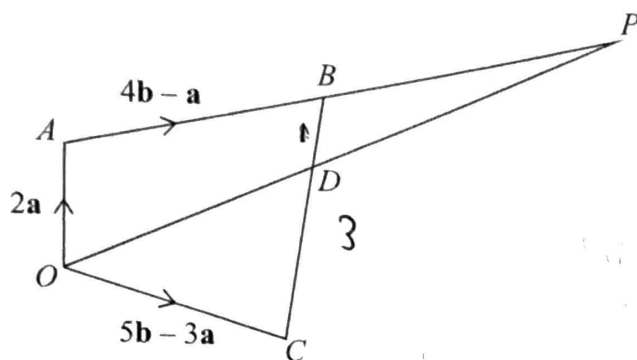


Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$.
Simplify your answer.

$$\vec{BC} = \vec{BA} + \vec{AO} + \vec{OC}$$

$$= -(4\mathbf{b} - \mathbf{a}) - 2\mathbf{a} + 5\mathbf{b} - 3\mathbf{a}$$

$$= -4\mathbf{b} + \mathbf{a} - 2\mathbf{a} + 5\mathbf{b} - 3\mathbf{a}$$

$$= 5\mathbf{b} - 4\mathbf{b} - \mathbf{a} - 3\mathbf{a}$$

$$= \underline{\underline{\mathbf{b} - 4\mathbf{a}}}$$

$$\mathbf{b} - 4\mathbf{a}$$

(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n



- 28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are $(0.2, 2.6)$

Work out the coordinates of B

Show your working clearly.